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Gaze detection and gaze cuing in Alzheimer's disease

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ABSTRACT

People with Alzheimer's disease (AD) show problems with social processing in tasks which require the understanding of others' mental states. However traditional social processing tasks are cognitively complex, which may influence the effects of AD. Less is known about how AD influences more basic aspects of social perception, such as the ability to decode eye gaze direction or follow the gaze of another. The current research assessed whether those with AD showed difficulty in both explicitly decoding subtle manipulations of gaze direction (Study 1), and reflexively following another's eye gaze (Study 2). Those with AD were more impaired than a matched control group when making explicit discrimination distinctions between direct and averted gaze. In contrast people with Alzheimer's disease performed comparably to a control group when following gaze. This pattern indicates that more automatic aspects of social perception such as gaze following are unaffected by AD. In contrast, more controlled processes such as deciding whether someone is looking towards you are impaired in AD. This has implications for socially engaging with other people and interpreting their focus of interest.

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1. Introduction

While Alzheimer's disease (AD) has traditionally been conceptualised in terms of declines in cognitive skills such as memory and attention, there is increasing evidence that AD can also impair social skills important in interpersonal interactions. Specifically, people with AD have more problems in social cue processing: the ability to decode information about mental states such as emotions, interest, intentions and beliefs from behaviour and demeanour. Considerable research indicates that people with AD have difficulty in understanding others' emotions (Klein-Koerkamp, Beaudoin, Baciu, & Hot, 2012) and beliefs (Sandoz, Démonet, & Fossard, 2014). These failures to read the social cues given by other people predict behavioural problems (Shimokawa et al., 2001) and poorer quality of life (Phillips, Scott, Henry, Mowat, & Bell, 2010) in people with AD, independent of assessments of basic cognitive functioning. However most of the studies to date have looked at the effects of AD in social cue tasks which are cognitive complex, and the main aim of our study is to explore whether AD also impairs performance on much simpler tasks of basic social cue perception: detecting and following others' gaze direction.

Most of the literature on dementia and social cue processing uses relatively complex tasks. For example many studies have explored the effects of AD on mental state understanding using 'theory of mind' tasks assessing the understanding of belief reasoning and particularly the ability to take others' perspectives and discern when they have a false belief about the state of the world (e.g. Fernandez-Duque, Baird, & Black, 2009). These tasks are cognitively complex, and so it is difficult to determine whether the effects of AD are primarily problems with the tricky reasoning and decision-making required in these tasks. Recent studies and reviews (Kemp, Despres, Sellal, & Dufour, 2012; Poletti, Enrici, & Adenzato, 2012; Sandoz et al., 2014; Shany-Ur et al., 2011) have concluded that AD causes poor performance on a range of tasks assessing different aspects of mental state understanding, and that the effects of AD are greatest when the task is most cognitively complex. For example false belief tasks, which require participants to inhibit their own viewpoint on the world to take another person's mistaken perspective, are performed poorly by people with AD (Fernandez-Duque et al., 2009). The reviews conclude that the effects of AD on these complex social reasoning tasks are likely secondary consequences of problems with executive function, attention and reasoning (e.g. Sandoz et al., 2014).

It is therefore important to explore further whether aspects of social understanding which are less cognitively complex are also affected by AD. According to Baron-Cohen (2005) the ability to process and interpret eye-gaze cues is an important precursor to

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higher level social processing. Processing of gaze cues involves the superior temporal sulcus (Engell & Haxby, 2007), and other key cortical regions (such as the temporo-parietal junction and ventromedial prefrontal cortex) which may underlie the effects of AD on social understanding (Poletti et al., 2012). There are multiple different components of gaze processing, with differing developmental trajectories and neural substrates (Shepherd, 2010). One key aspect of gaze processing is determining *gaze direction*. Deciding whether someone is looking towards or away from you, and in which direction they are looking, is a very important step in social communication (see Kleinke, 1986 for a review). A second key aspect is *following the gaze* of another person (also known as joint attention), where directional cues inform us where others are attending in the environment (Driver et al., 1999). These two aspects of gaze processing have different cognitive and neural components (Shepherd, 2010). While gaze following develops relatively early in childhood (<1 year) and is largely automatic, explicit discrimination develops much later (>3 years) and requires controlled cognition (Doherty, 2006; Doherty, Anderson, & Howieson, 2009).

However, little is known about how AD impacts on these basic gaze processing tasks. Given that gaze following is largely reflexive, we predict that there will be little effect of AD on this task. In contrast, the more controlled nature of explicit gaze direction decisions will result in AD-related deficits.

2. Study 1: Eye gaze detection in Alzheimer's disease

Gaze detection tasks require people to make decisions about whether someone else is looking towards or away from them, and to decide whether any averted gaze is to the right or left. Different neural systems are involved in processing information about direct and averted gaze (Shepherd, 2010), with detection of potentially threatening direct gaze more automatic compared to decisions about direction of averted gaze. Deciding where someone else's gaze is directed is important in social interaction and in understanding others' intentions. Normal adult aging can influence gaze processing. Healthy older adults are less accurate than young at decoding gaze direction (Slessor, Phillips, & Bull, 2008), and less likely to use information about gaze direction to influence their social decisions (Slessor, Phillips, & Bull, 2010). There is relatively little literature on the effects of AD on gaze detection. A couple of studies have investigated whether people with AD used information about which object a person was looking at to attribute object preferences (Castelli et al., 2011; Laisney et al., 2013). For example, from a picture showing someone was looking at an ice-cream rather than a chocolate it can be concluded that they preferred the ice cream. People with AD were less likely to make preference attributions in accordance with gaze direction. However, Laisney et al. (2013) concluded that these differences were not due to any problems in *detecting* gaze direction, but instead might relate to the preference-attribution aspect of the task. Also Castelli et al. (2011) reported that AD-related errors in the gaze preference task were mostly due to a failure to understand the rather complex task instructions.

Bediou et al. (2009) more directly investigated the effects of AD on the ability to categorize subtle variants in gaze. The stimuli were photographs of faces with gaze averted varying degrees to the left or right. They found no effect of AD on the ability to detect gaze direction. However, there are a number of reasons to question whether this is a definitive demonstration that AD does not impact on gaze detection. Firstly only 10 participants with AD were assessed, and the analysis was as part of a larger study including people with mild cognitive impairment and frontotemporal dementia. This may mean that effects of AD were not clear in the larger sample. Also, the task presented relatively gross differences in gaze which may have resulted in ceiling effects. In addition, they did not include a 'direct gaze' condition, and this is important as distinguishing between direct and averted gaze involves different cognitive and neural components compared to making left/right decisions (Shepherd, 2010). Errors of judgment in a social context which involve misinterpreting whether someone is looking at you or not may have particularly important consequences for social interaction. Including more subtle variants of gaze, and direct gaze stimuli, allows for more sensitive testing of any impact of AD in gaze detection.

The current study aims to investigate how AD impacts on the ability to discriminate between direct and averted gaze. Given that the cognitive load of this task is relatively low, and the previous results of Bediou et al. (2009) indicating no effect of AD on gaze processing it could be predicted that there should be no differences between people with AD and age-matched healthy controls on this task. However, previous arguments that gaze detection requires controlled social processing (Baron-Cohen, 2005; Doherty, 2006; Shepherd, 2010) leads to the opposite prediction, that participants in the AD group will be significantly less accurate than the controls in determining eye gaze direction.

2.1. Method

2.1.1. Participants

Two groups of participants were recruited. 24 older adult controls (10 male) ranging in age from 65 to 87 years (M = 74.00 years, SD = 5.43 years) recruited from the participant panel at the University of Aberdeen and reimbursed for their travel expenses. All control participants were tested at the University of Aberdeen. The clinical group were 24 people with AD (10 male) ranging in age from 65 to 89 years (M = 74.56, SD = 5.70). Those with AD were recruited from three sources: (1) the Department of Old Age Psychiatry at Royal Cornhill Hospital, Aberdeen (2) local Alzheimer Scotland groups and (3) registered volunteers from the Scottish Dementia Clinical Research Network. Regardless of source of recruitment, all participants with AD included in this research received their diagnosis from an old age psychiatrist based at Royal Cornhill Hospital, Aberdeen. All patients met the diagnostic criteria for "probable" AD as established by the National Institute of Aging working group (McKhann et al., 2011) and had Mini-Mental State Examination (MMSE; Folstein, Folstein, & McHugh, 1975) scores between 17 and 30, falling in the mild to moderately demented range. Some of the participants with AD were medicated, with the majority being prescribed more than one medicine. Prescriptions for medications included cholinesterase inhibitors and antidepressants in addition to treatment for physical illnesses such as hypertension. All those with AD were tested in their own home. Exclusion criteria for all participants included comorbidity of dementia subtypes (other than probable AD for the AD group), other neurological disorders, and history of alcohol or drug abuse.

A contrast sensitivity function (CSF) chart (Pelli, Robson, & Wilkins, 1988) was used to assess visual contrast sensitivity in both groups. CSF is a measure of how well the visual system can discriminate objects from their background. This aspect of visual perception could be important in making the fine-grained judgments necessary for discriminating gaze direction. The MMSE was used to determine the cognitive status of both groups. All members of the control group achieved a score higher than 24, which is the recommended cut off (Chayer, 2002). The age, education, CSF and Mini-Mental State Examination (MMSE; Folstein et al., 1975) scores for both groups were analysed (see Table 1). The groups were generally well matched: there were no significant group differences for age, education or CSF. Participants with AD scored significantly lower in terms of their MMSE scores than the

Table 1
Study 1: Control and AD participants' background information.

	Control group (n = 24)			AD group (n = 24)			*t*-test		Effect size
	M	*SD*	*range*	*M*	*SD*	*range*	*t* (46)	*p*	*d*
Age	74.00	5.43	63–87	74.56	5.70	63–89	−1.46	0.15	0.10
Education	13.25	3.07	9–20	12.96	4.16	9–25	−0.27	0.78	0.07
CSF	1.73	0.23	1.10–1.95	1.67	0.22	1.35–1.95	0.79	0.43	0.26
MMSE	29.33	0.70	28–30	23.17	2.97	17–30	9.89	<0.001	2.94

Note: CSF = Pelli Robson visual contrast sensitivity function test score.

control group, although we note that there was some overlap in scores: in our early dementia sample some of those with AD scored between 28 and 30 on the MMSE.

2.1.2. Stimuli and procedure

The images for the gaze detection task were selected from the Facial Expressions of Emotions: Stimuli and Test (FEEST; Young, Perrett, Calder, Sprengelmeyer, & Ekman, 2002), and were the same stimuli used to measure gaze detection as Slessor et al. (2008): please see that paper for examples of the stimuli used. These images were full face with only the direction of the eye gaze being manipulated using Adobe Photoshop, and each image was positioned centrally on the screen. Four actors (2 male & 2 female), each displaying a neutral facial expression were used. There were a total of 32 images in total (each image was 14 cm × 16 cm), 8 images for each of the four actors, consisting of 2 showing direct gaze, the averted conditions were presented once for each actor, these were 1 pixel averted right (0.13° from the direct gaze condition), 2 pixel averted right (0.25° from the direct gaze condition) and 3 pixel averted right (0.38° from direct). These averted gaze conditions were also represented as 1, 2 & 3 pixel averted left (also 0.13°, 0.25° & 0.38° from the direct gaze condition respectively). Note that we calculated visual angle from our stimuli based on established methods (http://osdoc.cogsci.nl/3.1/visualangle/).

The images were pseudo-randomised and presented to the participants at a viewing distance of approximately 45 cm (measured with a tape measure to ensure reasonable precision) on a 17″ screen of a laptop computer. Responses were collected using a Cedrus RB-530 response box (San Pedro, USA); three of the buttons were labelled 'left', direct' and 'right''. Participants were asked to decide in which direction the person on the screen was looking. It was made clear that if they thought the person was looking straight ahead they were to press the button labelled 'direct', and participants' understanding of this instruction was verified. There were a total of 8 trials for each of the gaze conditions, accuracy scores for the left and right conditions were collapsed across each of the gaze aversion levels resulting in 4 gaze conditions for analysis; direct gaze, 1 pixel averted, 2 pixel averted and 3 pixel averted.

2.1.3. Ethics

The methods for this study, and Study 2, were ethically approved by both the University of Aberdeen psychology research ethics committee, and the NHS North of Scotland Research Ethics.

2.2. Results

Descriptive statistics for the control and AD groups can be found in Table 2. The dependent variable was percentage accuracy for direct and gaze averted by 1, 2 or 3 pixels. To analyse group differences on the gaze detection task, a 4 (gaze direction: direct, averted by 1, 2, or 3 pixels) × 2 (Group Category: Control vs AD) mixed design ANOVA was used. There was a main effect of gaze direction, F (3,138) = 172.86, $p < 0.001$, $\eta_p^2 = 0.79$, with performance combined across the groups being least accurate on the 1 pixel condition (M = 23.82%) and most accurate on the 3 pixel averted (M = 88.40%). Performance for the whole sample on the 1 pixel averted condition (23.82%) was below chance level (i.e., 33.33%), but on other conditions performance was well above chance. The main effect of group was significant, F (1,46) = 9.57, $p < 0.01$, $\eta_p^2 = 0.17$, with the AD group (M = 63.17) being less accurate overall across all gaze conditions than the control group (M = 73.85). The interaction between group × gaze direction was not significant, F (3,138) = 0.43, $p = 0.73$, $\eta_p^2 = 0.009$, indicating that the effect of AD was relatively consistent across all conditions. Analyses revealed that the MMSE scores of the AD group significantly correlated with overall task performance on the gaze detection task, $r(24) = 0.59$, $p < 0.01$, but not for the control group, $r(24) = 0.01$, $p = 0.96$. There was also a significant correlation between the contrast sensitivity function scores of the AD group and task performance, $r(24) = 0.50$, $p < 0.001$. There was no significant correlation between these variables for the controls $r(24) = -0.31$, $p = 0.13$.

2.3. Discussion

The aim of study one was to investigate if AD affected the ability to detect subtle differences in eye gaze direction. The results of the current research showed a significant difference in accuracy: people with AD made significantly more errors overall. The interaction between group and gaze direction was not significant, this suggests that people with AD made significantly more errors for all gaze conditions regardless of degree of aversion (e.g. direct 1, 2 & 3 pixel averted conditions). These results contrast with those of Bediou et al. (2009) who found no difference between people with AD and controls on a gaze detection task, and this discrepancy likely reflects the larger sample and more sensitive test in the current study. We cannot directly compare the results for specific degrees of gaze aversion between the current study and Bediou et al. (2009), as different methods of establishing visual angle were used in the stimuli in the two papers. For the current stimuli (see Slessor et al., 2008 for more detail) we calculated visual angle subtended by the stimuli change, and reported this in a standard format. For Bediou et al. (2009: more detail is provided on their stimuli in Franck et al., 2002) it seems unlikely that the angles they report (from 5 to 30°) reflect visual angle subtended on the retina as usually assessed in psychophysics. Also the current study included discrimination of direct from averted gaze. Difficulties with perceiving gaze as direct or averted may hold consequences for social interaction, for example failure to see that someone is looking at you may mean missing cues from others keen to engage.

The current results show that people with AD can show impairments of social perception on even quite basic tasks such as discriminating gaze direction. In the group with AD, performance on the gaze detection task was correlated with MMSE scores and visual contrast sensitivity function. These correlations were only found in the AD group, suggesting that deficits in perception and cognition may influence the problems seen even in quite basic

Table 2
Mean percentage accuracy for the control and AD groups for each gaze condition.

	Control (n = 24)		AD (n = 24)	
	Mean	SD	Mean	SD
Direct Gaze	91.79	10.75	84.58	18.44
1 Pixel Averted	28.38	17.48	18.81	13.76
2 Pixel averted	80.96	18.67	66.33	26.32
3 Pixel Averted	93.83	11.61	82.98	23.71
Total	73.85	7.45	63.17	15.17

aspects of gaze processing. It is of interest to note that the overall group effect of AD on gaze detection is unlikely to be caused by visual perceptual difficulties more generally, given that there was no significant group difference in contrast sensitivity.

3. Study 2: Gaze cue following in Alzheimer's disease

In addition to accurately detecting the direction in which someone is looking, another important social skill is the ability to follow gaze direction to engage in joint attention. Although gaze cuing has not been explored explicitly in AD, one study has explored how mutual gaze behaviour is influenced by AD (Sturm et al., 2011). Participants were videotaped taking part in a 15 min conversation with a partner and the number of times the participant looked at their partner's eyes was coded from the footage. The frequency of eye contact was found by the coders to be comparable between the control participants and people with AD. This suggests that the tendency to look at eyes during social interaction is preserved in AD.

Tasks used to study gaze cuing in other populations (Driver et al., 1999; Slessor et al., 2008; Slessor, Laird, Phillips, Bull, & Fillippou, 2010; Slessor, Phillips, et al., 2010) have adapted an attention cueing paradigm (Posner, 1980). A face image is shown with gaze directed very strongly to the right or left. Then a target appears in either the location indicated by the gaze (congruent trials) or the opposite location (incongruent trials). Strong gaze following is indicated by much quicker reaction times to the target in congruent compared to incongruent trials. It is important to note that participants are not instructed to look at the gaze direction at all, and their task is to respond as quickly as possible to the target. The tendency to follow another's gaze is generally conceptualised as reflexive and automatic (Shepherd, 2010). Given that AD tends to impact most on tasks which require cognitive control and have a high attentional load, it can therefore be predicted that AD would likely have relatively little effect on a reflexive gaze cueing task. Although there have not been previous studies of gaze cuing in AD, a number of studies have used adapted attention tasks to look at the effects of AD of orientation of visual attention generally, using non-social stimuli. In these paradigms participants were asked to respond to a target after presentation of an arrow which pointed in a direction either congruent or incongruent with the target. Parasuraman, Greenwood, Haxby, and Grady (1992) found that people with AD had the same level of benefit as the age matched control group when responding to a target at a location cued by an arrow. This result has been supported by subsequent studies (e.g. Tales, Muir, Bayer, & Snowden, 2002).

However, it should be noted that in the course of normal aging, older adults show smaller gaze cueing effects compared to their younger counterparts (Slessor, Laird, et al., 2010; Slessor, Phillips, et al., 2010; Slessor et al., 2008, 2016). While the cause of these age differences is not yet understood, it is plausible that gaze cueing tasks might be more subject to controlled motivational factors than usually recognised. This opens the possibility that AD might further exaggerate the age-related changes in gaze cueing magnitude.

In the current study we look at the effects of AD on gaze cuing, using a social variant of the visual attention task. Given the findings that directional visual cue orientation remains intact in AD (Tales et al., 2002), and that gaze cuing appears to be a relatively automatic task, it can be predicted that people with AD should show similar gaze cuing effects compared to age-matched controls. This would provide a contrast with the results of Study 1 which indicated significant effects of AD on subtle gaze discrimination.

3.1. Method

3.1.1. Participants

A subset of the participants in study one also took part in study two. 15 older adults without AD (9 male) ranging in age from 65 to 87 years (M = 74.20 years, SD = 5.43 years) and 15 people with AD (9 male) ranging in age from 65 to 86 years (M = 74.93, SD = 5.70) completed study two. The other AD participants from study one did not take part because either they did not wish to complete a second study, or had problems in carrying out the task. Control participants were selected from the sample in study one who matched the AD participants in terms of age and education. Analysis of age, education, visual acuity, contrast sensitivity function and Mini-Mental State Examination (MMSE; Folstein et al., 1975) scores for both groups is shown in Table 3. Only one significant difference was found between the groups, with AD patients displaying a significantly lower MMSE scores.

3.1.2. Stimuli and procedure

Gaze cuing task: The stimuli were a subset of photographs taken from the original set used by Slessor et al. (2008). The photographs were greyscale images of four actors (2 male and 2 female), approximately 9 cm × 11 cm in size and depicted a neutral facial expression. Gaze direction of each actor was manipulated by 6 pixels to the left or right (0.76° from direct gaze: see http://osdoc.cogsci.nl/3.1/visualangle/ for the formula to calculate visual angle). In total there were 96 trials (64 congruent and 32 incongruent trials), following the predominant method in the gaze cuing literature where gaze is predictive in the majority of trials. Presentation of images was randomized and the position of the target (left or right) was counterbalanced. The images were presented on a 17″ Toshiba laptop computer, with participants sitting approximately 45 cm from the screen.

Participants were instructed that for each trial a fixation cross would appear in the centre of the screen (for 1000 ms), and when the cross disappeared a target (a large asterisk) would then appear on the left or right. Participants were told that a face would appear after the cross disappeared but this was to be ignored and they were to concentrate on responding to the asterisk target as quickly and as accurately as they could. Faces appeared for 200 ms, looking in a direction congruent or incongruent with the target, and the face disappeared at the onset of the target. The target remained present until a response was made. Responses were collected by a Cedrus RB-530 response box (San Pedro, USA) response box, one button was labelled 'left' and the other 'right'.

Table 3
Study 2: Control and AD participants' background information.

	Control Group (n = 15)			AD Group (n = 15)			*t*-test		Effect size
	M	*SD*	*range*	*M*	*SD*	*range*	*t* (46)	*p*	*d*
Age	74.20	5.56	67–87	74.93	5.70	65–86	−0.36	0.72	0.13
Education	13.53	2.77	10–20	14.20	4.54	9–23	−0.48	0.63	0.18
CSF	1.77	0.19	1.35–1.95	1.69	0.23	1.35–1.95	1.04	0.31	0.38
MMSE	29.40	0.63	28–30	23.87	2.97	17–29	7.05	<0.001	2.57

Note: CSF = Pelli Robson visual contrast sensitivity function test score.

Table 4
Mean reaction time and percentage of errors of the control and AD groups for each condition of the gaze cuing task.

Group	Congruent gaze				Incongruent gaze			
	Mean (ms)	SD	% Error	SD	Mean (ms)	SD	% Error	SD
Control	395.50	46.80	0.21	0.55	412.20	56.84	0.83	1.30
AD	591.40	223.99	0.21	0.55	612.43	223.47	0.10	0.40

3.2. Results

Due to the low level of errors made on the task (see Table 4), the dependent variable in each condition was the reaction time for correct trials (Slessor et al., 2008; Slessor, Laird, et al., 2010; Slessor, Phillips, et al., 2010). Median reaction times were calculated for each participant for both congruent and incongruent trials. In accordance with Slessor et al. (2008) these median RTs for the task conditions were then reciprocal-transformed to minimise outliers and normalize score distributions (Howell, 2006). Note that the descriptive statistics reported in the table are the mean group RTs in the congruent and incongruent conditions before data were transformed to reciprocals; however the analysis was undertaken post transformation using the reciprocal reaction times.

Group differences were analysed by a mixed design ANOVA with one within subject factor of gaze congruence (congruent vs. incongruent) and one between subjects factor of group (Control vs. AD). There was a significant main effect of gaze congruence, $F\ (1,28) = 20.97$, $p < 0.001$, $\eta_p^2 = 0.43$, with both groups showing significantly faster RTs for congruent ($M = 493$ ms) than incongruent trials ($M = 512$ ms). A main effect of group was also revealed, $F\ (1,28) = 23.48$, $p < 0.01$, $\eta_p^2 = 0.45$, with the AD group ($M = 602$ ms) being significantly slower overall in both the congruent and incongruent conditions compared to the control group ($M = 404$ ms). The cue congruency × group interaction was not significant, $F\ (1,28) = 0.37$, $p = 0.54$, $\eta_p^2 = 0.01$, suggesting that there was no group differences in the strength of the gaze congruency effect. There were no correlations between MMSE scores and the size of the congruency effect for those with AD, $r(15) = -0.22$, $p = 0.43$, or the control group, $r(15) = 0.01$, $p = 0.13$. The correlations between CSF and congruency effect were also non-significant for both groups, control group, $r(15) = -0.39$, $p = 0.15$ and AD group $r(15) = -0.09$, $p = 0.74$.

3.3. Discussion

The aim of Study 2 was to investigate potential effects of AD on a gaze cuing task. Results indicate that although slower in responding to a target than the controls, the AD group showed equivalent orienting response to gaze direction compared to controls. Robust gaze cuing effects were shown in both the AD participants and healthy older people, indicating that all participants were attending to the direction of eye gaze. The lack of interaction between cue congruence and group suggests that the ability to follow gaze was not affected by AD. This provides a clear contrast with the finding that healthy older adults show less gaze cuing than young (Slessor et al., 2008): indicating very different effects of normal and pathological aging processes on this aspect of social perception. Following multiple manipulations of the gaze-cueing task Slessor et al. (2016) argued that age differences in joint attention reflect specific difficulties in extracting social information from the eye region. The current results indicate that these difficulties are not further exacerbated by the presence of AD.

Results from Study 2 support the idea that gaze cuing mechanisms are relatively automatic, and unaffected by early-stage AD. These results are in line with the previous general attention literature in target detection which also found no differences in attention cuing between people with AD and controls when the cues to target location were non-social (arrows) (e.g. Faust & Balota, 1997; Tales et al., 2002). Here we extend those findings to social stimuli (eyes). The current findings also fit with evidence that participants with AD used mutual gaze cues during interactions with their spouse on a comparable level to control participants (Sturm et al., 2011). Although the current study has a relatively small sample of participants with AD (n = 15) the number is comparable to other studies looking at effects of AD using a similar non-social cuing paradigm (Parasuraman et al., 1992; Tales et al., 2002), and other studies showing significant effects of AD on other social cognition tasks (Castelli et al., 2011; Laisney et al., 2013).

4. General discussion

The current research was the first to explore the effect of AD on discriminating between direct and averted gaze direction, and the ability to orient visual attention to gaze cues. Although previous research has found social cognition skills are poorer in those with AD than typically aging individuals when assessed using more complex social processing tasks (e.g. Fernandez-Duque et al., 2009; Laisney et al., 2013), the current study indicates that AD has a differential impact on eye gaze processing abilities. Determining if the gaze of others is direct or averted was found to be more difficult for those with AD (Study 1) whereas the ability to follow gaze cues was found to be preserved (Study 2).

Inconsistencies in performance between discrimination of gaze direction and following gaze in children has been suggested to be the result of two distinct gaze processing systems (Doherty, 2006). Doherty's two system theory may be used to explain the pattern of results in the current study. For example in AD, the learned component underpinning gaze discrimination may deteriorate earlier, resulting in a poorer ability to make judgments about direct and averted gaze. In contrast the reflexive system for gaze cuing is less affected by AD, allowing attention to be orientated

at the same level as the controls. The finding in the current research that MMSE scores were correlated with gaze detection but not gaze following in the AD group may be tentative evidence for eye gaze detection being more cognitively demanding than the gaze following task. It would be useful in future research to look in more detail at the cognitive components of gaze decoding in AD, using a wider variety of stimuli manipulations such as stimulus-onset-asynchrony to alter the influence of cognitive control processes in performance.

In addition to the correlation between MMSE and gaze detection performance, the group with AD also demonstrated a correlation between contrast sensitivity and task performance for gaze discrimination only. This suggests that the perceptual ability to differentiate the eyes may have also contributed to task performance. Also, we should note that some of the testing was carried out in participant's homes and this might have influenced results for those tested at home due to less control of experimental factors such as exact placing of the laptop for testing, lighting conditions, and distraction from the environment. We did carefully measure the distance from laptop screen to participant, and had good co-operation from the participants in their own homes and their families to minimise distraction. While greater experimental control could be achieved by testing all participants in the same laboratory, this would also have excluded some people from taking part who would have difficulty in travelling. We also note that while we equated distance from the screen for each participant, without the use of a chinrest participants will have moved to some extent, and this would impact upon the visual angle of the stimuli. Future studies could consider controlling for this.

One caveat in interpreting the current pattern of impaired gaze detection but preserved gaze cuing in AD is that there were different sample sizes in the two studies. It seems unlikely that the smaller sample size in Study 2 can explain the absence of group effect though, because other studies investigating the effects of AD on social cognitive tasks generally produce medium effect sizes in samples sizes identical to that in our Study 2 (e.g. Castelli et al., 2011; Laisney et al., 2013).

We also do not have direct evidence of how brain changes in AD influence the pattern of results here. Different cognitive and neural mechanisms may be involved in gaze detection and gaze following (Shepherd, 2010), and complex cortical networks involving multiple temporal and frontal lobe systems may deteriorate in mild-moderate AD, influencing social perception (Poletti et al., 2012). It is an important goal for future research to understand more about the link between brain changes in cortical dementias such as AD and the range of social perception and more complex reasoning and decision-making tasks involved in interpersonal understanding. An important goal for future research is to explore, in a single sample of people with AD, a larger battery of (1) more basic and reflexive social processing tasks (such as gaze cueing and visual search) to assess social attention, and (2) tasks such as emotion labelling and false belief reasoning to tap into higher-order understanding of mental states. Looking at the effects of AD on a wide range of tasks in the same sample would also allow modelling of how these different types of social process are inter-related. Looking at a sample which was also well characterised in terms of cognitive assessment and neuroimaging would also allow better models of the cognitive and neural components of social processing changes (and stability) in AD.

The difficulties that those with AD demonstrated on the gaze detection task adds to evidence of difficulty decoding subtle social cues in dementia. For example, there is evidence for difficulties in decoding emotion effectively in AD (see Klein-Koerkamp et al., 2012 for a meta-analysis). The ability to determine direct gaze appears to hold particular salience in the area of social cognition. For example in younger adulthood, direct gaze has been found previously to augment perceptions of emotions such as anger and happiness where averted gaze increases perception of fear and sadness (Adams & Kleck, 2003, 2005). Faces with direct gaze have also been perceived to be more likeable than faces with averted gaze (Mason, Tatkow, & Macrae, 2005). If people with AD are unable to identify and process the direct eye gaze of another person effectively this may hold consequences for the manner in which they engage socially and their subjective perception of the person with whom they are engaging.

4.1. Conclusions

People with AD were significantly poorer at decoding eye gaze direction compared to age-matched controls, whereas they showed comparable performance to controls when following gaze cues. Differential effects for gaze detection and gaze following may be attributable to differences in the nature of these two abilities, such that gaze detection is a learned cognitive skill, whereas gaze following is a more innate and reflexive process. However, given the importance of detecting gaze direction for many aspects of social interaction (Argyle & Dean, 1965; Kleinke, 1986) declines in this ability may have consequences for social function and the psychological well-being of those living with a diagnosis and their families.

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